

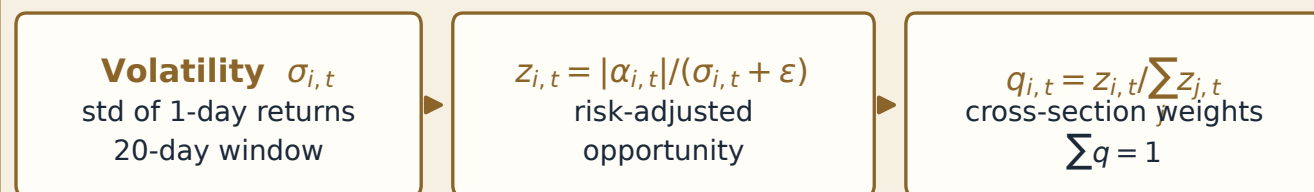
1 Forecast

Point-in-time market and SEC features → XGBoost predicts 20-day active return



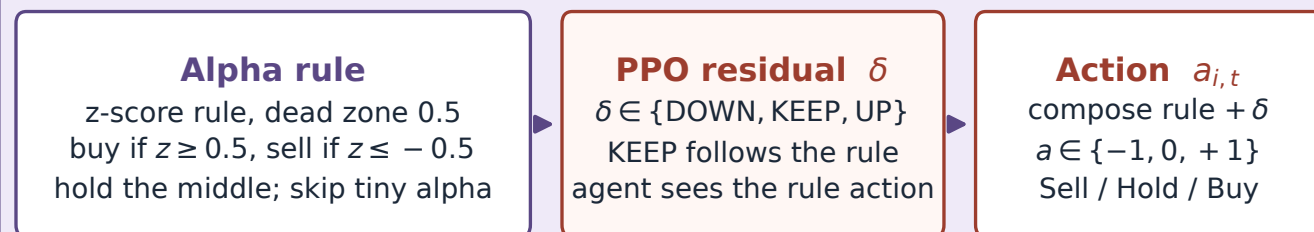
2 Risk adjustment

Rolling 20-day volatility scales signal strength; q later sizes the trade



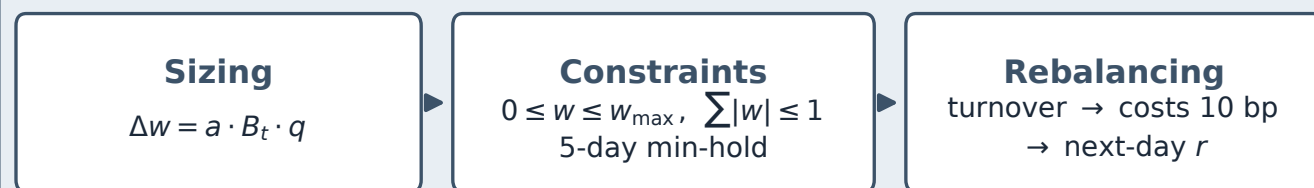
3 Decision — residual PPO (default)

Rule proposes Buy/Hold/Sell; PPO outputs a residual around that rule, not a raw action



4 Position sizing, constraints, rebalancing

Incremental default: $\Delta w_{i,t} = a_{i,t} B_t q_{i,t}$ with $B_t = 0.2$, $w_{\max} = 0.2$, long-only



5 Reward

$R_{t+1} = \log(V_{t+1}/V_t) - \lambda_{\text{trade}} \text{Turnover}_t + \lambda_{\text{align}} \sum a_{i,t} \text{sign}(\alpha_{i,t}) q_{i,t}$
Costs already in V ; $\lambda_{\text{trade}} = 10 \text{ bp}$, $\lambda_{\text{align}} = 5 \text{ bp}$.